Customer Segmentation using RFM Analysis

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Problem Statement

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E-commerce businesses generate vast amounts of transactional data but often struggle to leverage it for actionable customer segmentation. Understanding purchasing patterns is crucial for predicting behavior, improving retention, reducing churn, and maximizing customer lifetime value. A structured approach is needed to identify high-value customers, detect trends, and optimize targeted marketing strategies.

Objective:

- Utilize the RFM (Recency, Frequency, Monetary) framework to segment e-commerce customers.
- Analyze purchasing patterns to identify high-value customers and at-risk segments.
- Create targeted marketing strategies to improve customer retention and satisfaction.

Dataset Overview

- **Source:** [Kaggle - E-commerce Dataset]
- **Size:** (541909, 8)
- **Timeframe:** 2010-12-01 - 2011-12-09

Key Variables:

- **InvoiceNo:** Unique transaction ID
- **StockCode:** Unique product identifier
- **Description:** Product description
- **Quantity:** Number of items purchased
- **InvoiceDate:** Timestamp of purchase
- **UnitPrice:** Price per unit in GBP
- **CustomerID:** Unique customer identifier
- **Country:** Country of the customer

Data Cleaning and Preparation

- Handled Missing Values
- Handled duplicates
- Handled cancelled products
- Handled non product StockCodes

1. Handled Missing Values

2. Handled duplicates

3. Handled cancelled products

4. Handled non product stockcodes

```
[8]: df.isna().sum()

[8]: InvoiceNo           0
     StockCode           0
     Description      1454
     Quantity            0
     InvoiceDate         0
     UnitPrice           0
     CustomerID     135080
     Country             0
     dtype: int64
```

- CustomerID has around 135000 null values which is around 24% of data. Since our analysis will include segmenting customers into different groups, it is better to remove the null CustomerID.
- Removing null CustomerID also removed null Descriptions.

1. Handled Missing Values

2. Handled duplicates

3. Handled cancelled products

4. Handled non product stockcodes

```
df.duplicated().sum()
```

```
np.int64(5225)
```

```
df.drop_duplicates(inplace=True)
```

- There were 5225 duplicates in the dataset.
- These were removed.

1. Handled Missing Values

2. Handled duplicates

3. Handled cancelled products

4. Handled non product StockCodes

```
df.query('Quantity < 0')
```

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
141	C536379	D	Discount	-1	2010-01-12 09:41:00	27.50	14527.0	United Kingdom
154	C536383	35004C	SET OF 3 COLOURED FLYING DUCKS	-1	2010-01-12 09:49:00	4.65	15311.0	United Kingdom
235	C536391	22556	PLASTERS IN TIN CIRCUS PARADE	-12	2010-01-12 10:24:00	1.65	17548.0	United Kingdom
236	C536391	21984	PACK OF 12 PINK PAISLEY TISSUES	-24	2010-01-12 10:24:00	0.29	17548.0	United Kingdom
237	C536391	21983	PACK OF 12 BLUE PAISLEY TISSUES	-24	2010-01-12 10:24:00	0.29	17548.0	United Kingdom
...	...	...	...	...	...	...	...	...
540449	C581490	23144	ZINC T-LIGHT HOLDER STARS SMALL	-11	2011-09-12 09:57:00	0.83	14397.0	United Kingdom
541541	C581499	M	Manual	-1	2011-09-12 10:28:00	224.69	15498.0	United Kingdom
541715	C581568	21258	VICTORIAN SEWING BOX LARGE	-5	2011-09-12 11:57:00	10.95	15311.0	United Kingdom
541716	C581569	84978	HANGING HEART JAR T-LIGHT HOLDER	-1	2011-09-12 11:58:00	1.25	17315.0	United Kingdom
541717	C581569	20979	36 PENCILS TUBE RED RETROSPOT	-5	2011-09-12 11:58:00	1.25	17315.0	United Kingdom

8872 rows × 8 columns

- There were 8872 rows for which quantity is negative. This is indicate: errors in data entry or cancelled products or returned products.
- The InvoiceNo of all these products start with ‘C’, which could indicate “Cancelled” orders.
- These were removed from the dataset.

1. Handled Missing Values

2. Handled duplicates

3. Handled cancelled products

4. Handled non product StockCode

StockCodes	Description
POST	POSTAGE
C2	CARRIAGE
M	MANUAL
DOT	DOTCOM POSTAGE
BANK CHARGES	BANK CHARGES

- There are certain StockCodes which do not belong to any products.
- All the rows containing such StockCodes were removed.

Exploratory Data Analysis

Daily Sales Analysis

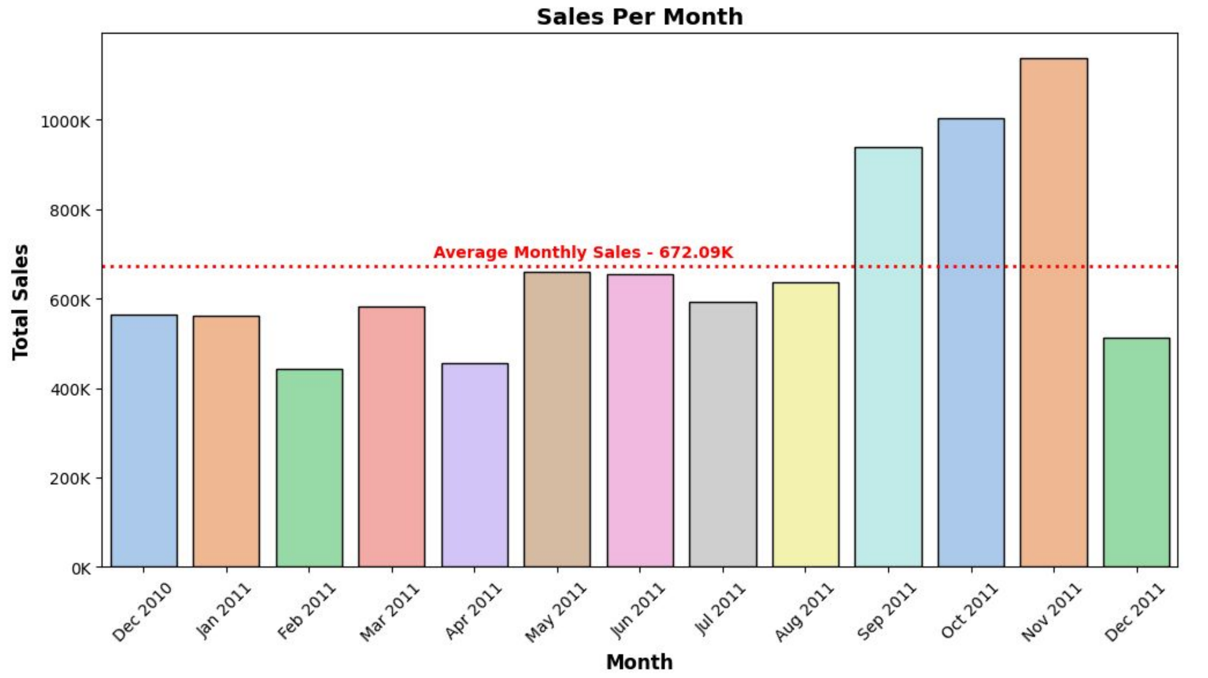
- Sales are concentrated on few peak days
- Strong year-end seasonality, with most top sales days occurring in Q4.
- Daily sales are right-skewed, a few days dominate overall revenue.

```
daily_sales.sort_values(['TotalSales'], ascending=False).head(10)
```

	InvoiceDate	TotalSales
304	2011-12-09	184170.66
235	2011-09-20	103280.33
32	2011-01-18	87065.81
248	2011-10-05	73396.57
290	2011-11-23	70135.74
231	2011-09-15	69520.59
302	2011-12-07	68575.60
279	2011-11-10	68098.55
196	2011-08-04	61599.77
278	2011-11-09	60760.33

Monthly Sales Analysis

- Average Monthly Sale is 672.09K
- Majority of months fall below average.
- There is sharp increase in sales in Sep, Oct and Nov.
- Dec show volatility, decreasing sharply after Nov peak.



Revenue growth is seasonal and Q4-driven.

26%

Customer contribute to 80% of Revenue

21%

Product contribute to 80% of Revenue

RFM ANALYSIS & CUSTOMER SEGMENTATION

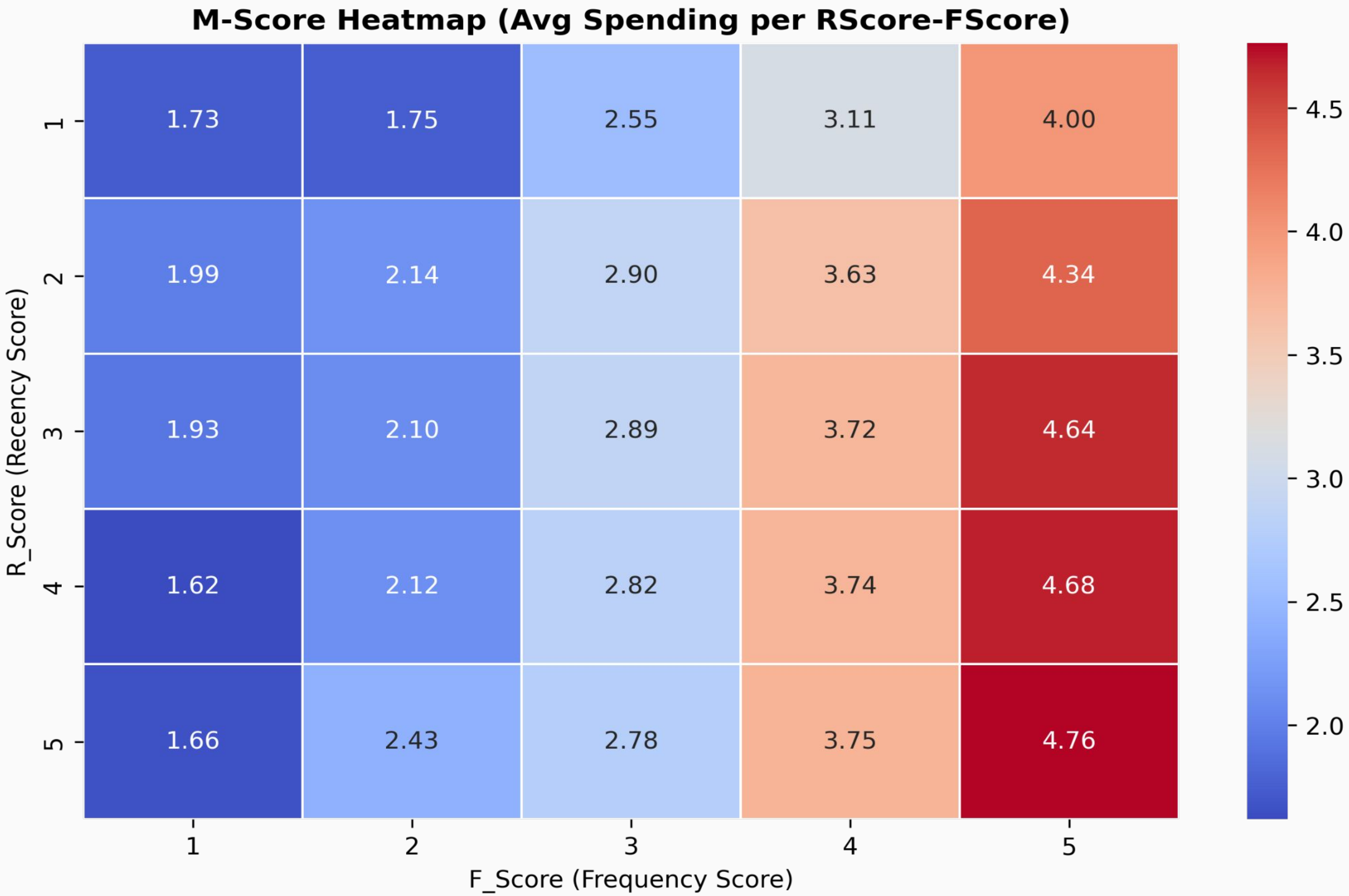
WHAT IS RFM AND CUSTOMER SEGMENTATION?

What is RFM?

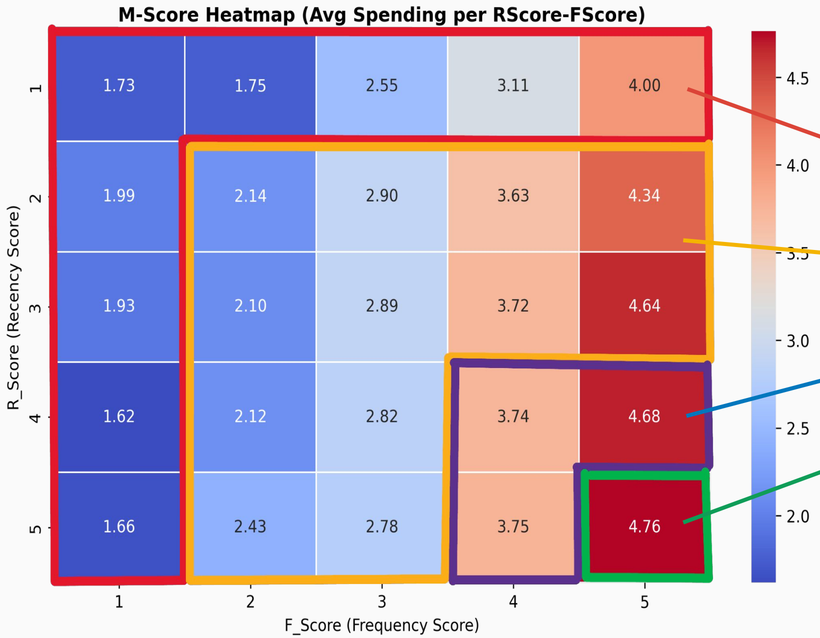
Recency (R): Days since last purchase
Frequency (F): Number of purchases
Monetary (M): Total spending

Customers are segmented into **five equal buckets** based on Recency, Frequency, and Monetary values. Each customer is ranked for each metric, assigned a **score from 1 to 5**, and their scores are summed to derive an overall **RFM score** for analysis.

HEATMAP BASED ON RFM SCORE



RFM Analysis and Customer Segmentation



Based on this RFM Score analysis, we have 4 types of customers.

1. **Dormant Customers**
2. **At Risk Customers**
3. **Loyal Customers**
4. **High Value Customers**

DASHBOARD

Country: All

Month: All

Customer Segmentation Dashboard (RFM Analysis)

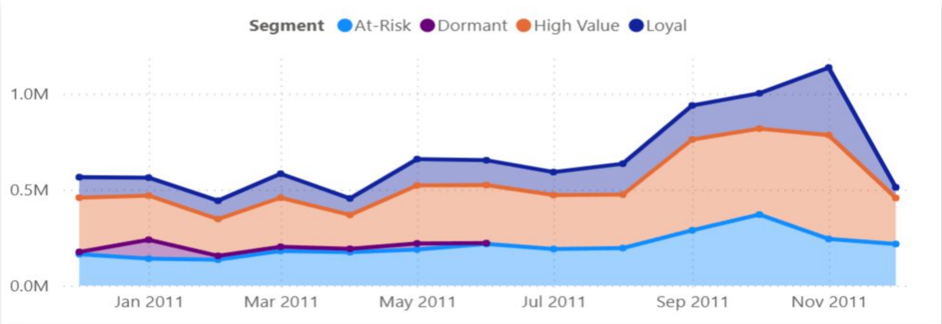
Qtr 1	Qtr 2	Qtr 3	Qtr 4	2010	2011

8.74M
TotalRevenue

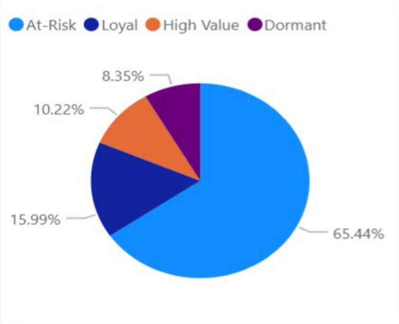
4335
Active Customers

5M
Sales Volume

Revenue Contributions by Segment



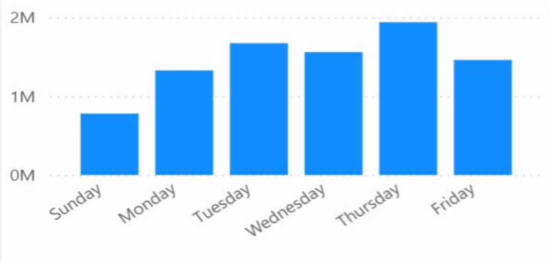
Customers by Segment



RFM Heatmap

RScore	1	2	3	4	5
1	362	307	138	47	13
2	199	199	227	163	50
3	165	182	188	208	117
4	97	121	177	249	244
5	44	58	137	200	443

Revenue by Days Of Week



Active Customers Over Time

Top/Bottom Customers

CustomerID	Total Sales	Sales Volume
14646	2,79,138.02	197420
18102	2,59,657.30	64124
17450	1,94,390.79	69973
16446	1,68,472.50	80997
14911	1,36,161.83	80404
12415	1,24,564.53	77669
14156	1,16,560.08	57755
17511	91,062.38	64549
Total	**87,37,227.64**	**5155661**

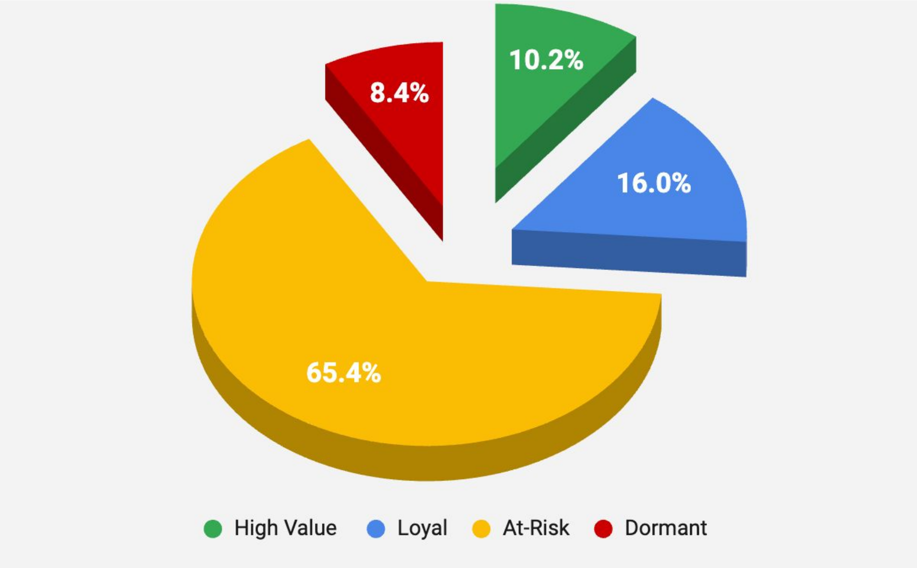
Recommendations

HIGH VALUE CUSTOMERS

Characteristics: Brand advocates with exceptional engagement and spending.

Behaviour: Low Recency, High Frequency, High Monetary.

Strategy: Reward with loyalty programs and exclusives.



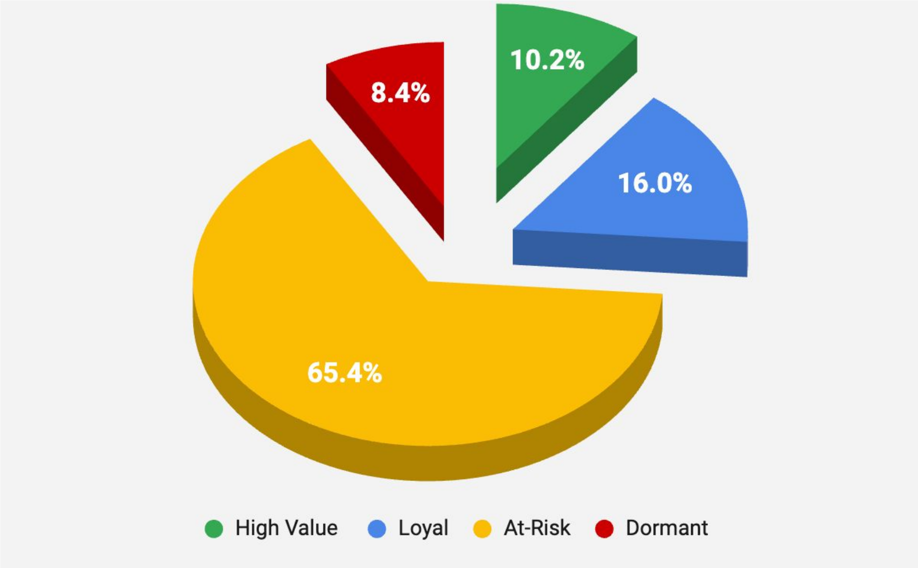
Recommendations

LOYAL CUSTOMERS

Characteristics: Consistent buyers with moderate activity.

Behaviour: Average Recency (recent purchases), Moderately High Frequency, High Monetary.

Strategy: Upsell/Cross-sell opportunities to boost value.



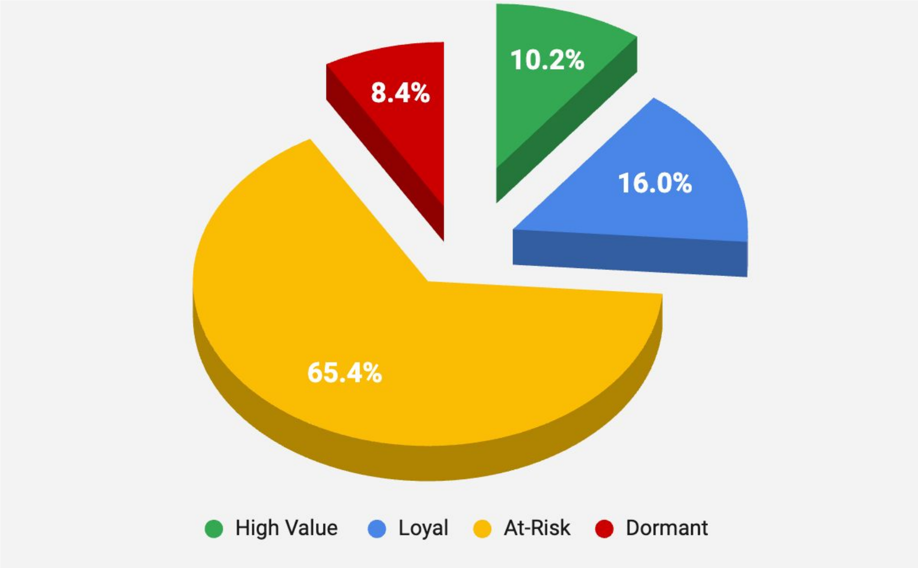
Recommendations

AT-RISK CUSTOMERS

Characteristics: Not-so Consistent buyers

Behaviour: Moderately High Recency, Low Frequency, Variable Monetary.

Strategy: Personalized emails and Limited-time promotions to regain interest.



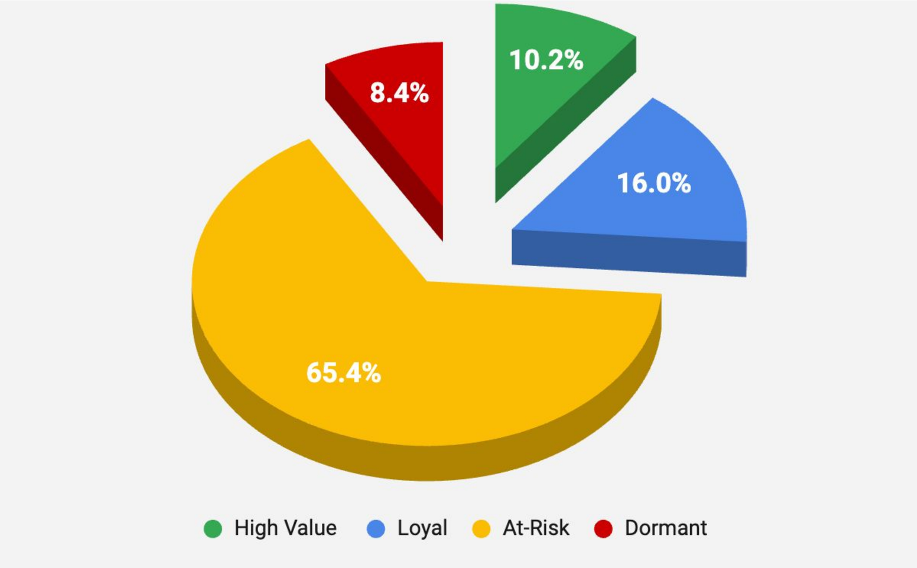
Recommendations

DORMANT CUSTOMERS

Characteristics: Declining engagement with potential churn risk.

Behaviour: High Recency, Below Average Frequency, Low Monetary.

Strategy: Win-back campaigns or surveys to address dissatisfaction.



THANK YOU